

1D DFT in Matrix Notation

With matrix notation instead of summation, this question helps you understand 1D DFT as a change of basis in a finite-dimensional vector space.

Consider the space of sequences of 4 complex numbers, $\{h_0, h_1, h_2, h_3\}$. It can be considered as a 4-dimensional complex vector space

$$\vec{h} = \begin{pmatrix} h_0 \\ h_1 \\ h_2 \\ h_3 \end{pmatrix}$$

with addition defined by

$$\vec{h} + \vec{g} = \begin{pmatrix} h_0 \\ h_1 \\ h_2 \\ h_3 \end{pmatrix} + \begin{pmatrix} g_0 \\ g_1 \\ g_2 \\ g_3 \end{pmatrix} = \begin{pmatrix} g_0 + h_0 \\ g_1 + h_1 \\ g_2 + h_2 \\ g_3 + h_3 \end{pmatrix}$$

and (complex) scalar multiplication defined by

$$c\vec{h} = \begin{pmatrix} ch_0 \\ ch_1 \\ ch_2 \\ ch_3 \end{pmatrix}$$

The zero vector of this vector space is

$$\vec{0} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

and the 4 standard basis vectors are

$$\vec{e}_0 = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \vec{e}_1 = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \vec{e}_2 = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \vec{e}_3 = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

Because $\vec{h} = \vec{0}$ if and only if $h_0 = h_1 = h_2 = h_3 = 0$, every vector can be written in a unique way as a linear combination

$$\vec{h} = \begin{pmatrix} h_0 \\ h_1 \\ h_2 \\ h_3 \end{pmatrix} = h_0\vec{e}_0 + h_1\vec{e}_1 + h_2\vec{e}_2 + h_3\vec{e}_3$$

The vector space can be made an inner product space by defining the inner product

$$\langle \vec{h}, \vec{g} \rangle = h_0\overline{g_0} + h_1\overline{g_1} + h_2\overline{g_2} + h_3\overline{g_3}$$

(a) Show that

$$\langle \vec{g}, \vec{h} \rangle = \overline{\langle \vec{h}, \vec{g} \rangle}$$

The norm (length) of a vector $\vec{h}$ can be defined to be

$$|\vec{h}| = \sqrt{\langle \vec{h}, \vec{h} \rangle}$$

It can be checked easily that

$$\langle \vec{e}_i, \vec{e}_j \rangle = \delta_{ij}$$

where $i, j = 0, 1, 2, 3$.

Now consider the 4 vectors

$$\vec{e}'_k = \begin{pmatrix} e^{-2\pi i k \cdot 0/4} \\ e^{-2\pi i k \cdot 1/4} \\ e^{-2\pi i k \cdot 2/4} \\ e^{-2\pi i k \cdot 3/4} \end{pmatrix} = \begin{pmatrix} 1 \\ e^{-i\pi k/2} \\ e^{-i\pi k} \\ e^{-3i\pi k/2} \end{pmatrix}$$

for $k = 0, 1, 2, 3$. Or explicitly,

$$\vec{e}'_0 = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}$$

$$\vec{e}'_1 = \begin{pmatrix} 1 \\ e^{-i\pi/2} \\ e^{-i\pi} \\ e^{-3i\pi/2} \end{pmatrix}$$

$$\vec{e}'_2 = \begin{pmatrix} 1 \\ e^{-i\pi} \\ e^{-2i\pi} \\ e^{-3i\pi} \end{pmatrix}$$

$$\vec{e}'_3 = \begin{pmatrix} 1 \\ e^{-3i\pi/2} \\ e^{-3i\pi} \\ e^{-9i\pi/2} \end{pmatrix}$$

(b) Show that

$$\langle \vec{e}'_k, \vec{e}'_l \rangle = 4\delta_{kl}$$

Define

$$F = ((\vec{e}'_0) \quad (\vec{e}'_1) \quad (\vec{e}'_2) \quad (\vec{e}'_3)) = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & e^{-i\pi/2} & e^{-i\pi} & e^{-3i\pi/2} \\ 1 & e^{-i\pi} & e^{-2i\pi} & e^{-3i\pi} \\ 1 & e^{-3i\pi/2} & e^{-3i\pi} & e^{-9i\pi/2} \end{pmatrix}$$

that is

$$F_{nk} = e^{-2\pi i kn/4}$$

Note that F is symmetric.

(c) Show that in the representation of the standard basis,

$$F\vec{e}_i = \vec{e}'_i$$

for $i = 0, 1, 2, 3$.

Also define the conjugate transpose of matrix F by

$$F^* = \overline{F^T}$$

(d) Find F^*F by (b). Hint: Note that the rows of F^T are the row basis vectors

$$F^T = \begin{pmatrix} (\vec{e}'_0)^T \\ (\vec{e}'_1)^T \\ (\vec{e}'_2)^T \\ (\vec{e}'_3)^T \end{pmatrix}$$

(e) Find FF^* by (d) and the symmetry of F .

Now consider the representation of $\vec{h}$ in the two basis

$$\vec{h} = h_0\vec{e}_0 + h_1\vec{e}_1 + h_2\vec{e}_2 + h_3\vec{e}_3 = H_0\vec{e}'_0 + H_1\vec{e}'_1 + H_2\vec{e}'_2 + H_3\vec{e}'_3$$

and hence in matrix form

$$\begin{pmatrix} h_0 \\ h_1 \\ h_2 \\ h_3 \end{pmatrix} = H_0 F \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} + H_1 F \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} + H_2 F \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} + H_3 F \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} = F \begin{pmatrix} H_0 \\ H_1 \\ H_2 \\ H_3 \end{pmatrix}$$

(f) Obtain H_0, H_1, H_2, H_3 in matrix form by (d) and/or (e).

Consider the sequence $\{1, 0, i, 0\}$.

(g) Find H_0, H_1, H_2, H_3 .

(h) Using the answer in (g), verify that the Parseval's theorem holds.

(a) By comparison

$$\begin{aligned}\langle \vec{h}, \vec{g} \rangle &= h_0 \overline{g_0} + h_1 \overline{g_1} + h_2 \overline{g_2} + h_3 \overline{g_3} \\ \langle \vec{g}, \vec{h} \rangle &= g_0 \overline{h_0} + g_1 \overline{h_1} + g_2 \overline{h_2} + g_3 \overline{h_3}\end{aligned}$$

Obviously $\langle \vec{g}, \vec{h} \rangle = \overline{\langle \vec{h}, \vec{g} \rangle}$ because $\overline{z_1 z_2} = \overline{z_1} \cdot \overline{z_2} = \overline{z_1} \cdot z_2$ for any complex number z_1, z_2 .

(b)

$$\begin{aligned}\langle \vec{e}'_k, \vec{e}'_l \rangle &= \begin{pmatrix} 1 & e^{-\frac{i\pi k}{2}} & e^{-i\pi k} & e^{-\frac{3i\pi k}{2}} \end{pmatrix} \begin{pmatrix} 1 \\ e^{\frac{i\pi l}{2}} \\ e^{i\pi l} \\ e^{\frac{3i\pi l}{2}} \end{pmatrix} \\ &= 1 + e^{\frac{i\pi(l-k)}{2}} + e^{i\pi(l-k)} + e^{\frac{3i\pi(l-k)}{2}} \\ &= 1 + i^{(l-k)} + (-1)^{(l-k)} + (-i)^{(l-k)}\end{aligned}$$

When $l - k = 0$, $\langle \vec{e}'_k, \vec{e}'_l \rangle = 1 + 1 + 1 + 1 = 4$.

When $l - k = 1$ or -3 , $\langle \vec{e}'_k, \vec{e}'_l \rangle = 1 + i + (-1) + (-i) = 0$.

When $l - k = 2$ or -2 , $\langle \vec{e}'_k, \vec{e}'_l \rangle = 1 + (-1) + 1 + (-1) = 0$.

When $l - k = 3$ or -1 , $\langle \vec{e}'_k, \vec{e}'_l \rangle = 1 + (-i) + (-1) + i = 0$.

(c) Note that

$$((\vec{e}_0) \quad (\vec{e}_1) \quad (\vec{e}_2) \quad (\vec{e}_3)) = I = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

So it is obvious that

$$F ((\vec{e}_0) \quad (\vec{e}_1) \quad (\vec{e}_2) \quad (\vec{e}_3)) = FI = F = ((\vec{e}'_0) \quad (\vec{e}'_1) \quad (\vec{e}'_2) \quad (\vec{e}'_3))$$

(d)

$$\begin{aligned}F^* F &= \begin{pmatrix} (\vec{e}'_0)^T \\ (\vec{e}'_1)^T \\ (\vec{e}'_2)^T \\ (\vec{e}'_3)^T \end{pmatrix} ((\vec{e}'_0) \quad (\vec{e}'_1) \quad (\vec{e}'_2) \quad (\vec{e}'_3)) \\ &= \begin{pmatrix} \langle \vec{e}'_0, \vec{e}'_0 \rangle & \langle \vec{e}'_1, \vec{e}'_0 \rangle & \langle \vec{e}'_2, \vec{e}'_0 \rangle & \langle \vec{e}'_3, \vec{e}'_0 \rangle \\ \langle \vec{e}'_0, \vec{e}'_1 \rangle & \langle \vec{e}'_1, \vec{e}'_1 \rangle & \langle \vec{e}'_2, \vec{e}'_1 \rangle & \langle \vec{e}'_3, \vec{e}'_1 \rangle \\ \langle \vec{e}'_0, \vec{e}'_2 \rangle & \langle \vec{e}'_1, \vec{e}'_2 \rangle & \langle \vec{e}'_2, \vec{e}'_2 \rangle & \langle \vec{e}'_3, \vec{e}'_2 \rangle \\ \langle \vec{e}'_0, \vec{e}'_3 \rangle & \langle \vec{e}'_1, \vec{e}'_3 \rangle & \langle \vec{e}'_2, \vec{e}'_3 \rangle & \langle \vec{e}'_3, \vec{e}'_3 \rangle \end{pmatrix} \\ &= \begin{pmatrix} 4 & 0 & 0 & 0 \\ 0 & 4 & 0 & 0 \\ 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & 4 \end{pmatrix}\end{aligned}$$

(e) Since F is symmetric, $F = F^T$. Then we have

$$FF^* = F\overline{F^T} = F\overline{F} = \overline{(\overline{F}F)} = \overline{(F^T F)} = \overline{F^* F} = \begin{pmatrix} 4 & 0 & 0 & 0 \\ 0 & 4 & 0 & 0 \\ 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & 4 \end{pmatrix}$$

(f) By

$$\begin{aligned} F \begin{pmatrix} H_0 \\ H_1 \\ H_2 \\ H_3 \end{pmatrix} &= \begin{pmatrix} h_0 \\ h_1 \\ h_2 \\ h_3 \end{pmatrix} \\ F^* F \begin{pmatrix} H_0 \\ H_1 \\ H_2 \\ H_3 \end{pmatrix} &= \begin{pmatrix} 4 & 0 & 0 & 0 \\ 0 & 4 & 0 & 0 \\ 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & 4 \end{pmatrix} \begin{pmatrix} H_0 \\ H_1 \\ H_2 \\ H_3 \end{pmatrix} = 4 \begin{pmatrix} H_0 \\ H_1 \\ H_2 \\ H_3 \end{pmatrix} = F^* \begin{pmatrix} h_0 \\ h_1 \\ h_2 \\ h_3 \end{pmatrix} \\ \begin{pmatrix} H_0 \\ H_1 \\ H_2 \\ H_3 \end{pmatrix} &= \frac{F^*}{4} \begin{pmatrix} h_0 \\ h_1 \\ h_2 \\ h_3 \end{pmatrix} \end{aligned}$$

(g) Given $(h_0, h_1, h_2, h_3) = \{1, 0, i, 0\}$

$$\begin{pmatrix} H_0 \\ H_1 \\ H_2 \\ H_3 \end{pmatrix} = \frac{F^*}{4} \begin{pmatrix} 1 \\ 0 \\ i \\ 0 \end{pmatrix} = \frac{1}{4} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & e^{i\pi/2} & e^{i\pi} & e^{3i\pi/2} \\ 1 & e^{i\pi} & e^{2i\pi} & e^{3i\pi} \\ 1 & e^{3i\pi/2} & e^{3i\pi} & e^{9i\pi/2} \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ i \\ 0 \end{pmatrix} = \frac{1}{4} \begin{pmatrix} 1+i \\ 1-i \\ 1+i \\ 1-i \end{pmatrix}$$

(h)

$$\begin{aligned} \|\vec{h}\|^2 &= \langle \vec{h}, \vec{h} \rangle = 1^2 + 0^2 + 1^2 + 0^2 = 2 \\ \sum |\vec{H}|^2 &= \frac{1}{4} (4 \times (1+i)(1-i)) = 2 \end{aligned}$$

So Parseval's theorem holds.

This question helps you to understand with a simple example why the response in a Linear-Time-Invariant discrete time system is a discrete linear convolution.

The impulse at $n = 0$ of a discrete time system is defined by

$$\delta[n] = \delta_{n0}$$

Therefore, the general impulse at $n = n_0$ is

$$\delta[n - n_0] = \delta_{n-n_0,0} = \delta_{nn_0}$$

Let the finite-impulse response (FIR) of $\delta[n]$ be

$$h[n] = \begin{cases} 1 & n = 0,1,2 \\ 0 & \text{Otherwise} \end{cases}$$

----- Eq. *

Now consider the input signal

$$g[n] = \begin{cases} 1 & n = 2,3 \\ 2 & n = 4,5 \\ 0 & \text{Otherwise} \end{cases}$$

(a) Write $g[n]$ explicitly as the linear combination of some impulses.

Assume that the system is time-invariant. In other words, it responds in the same way with respect to an input impulse regardless of the time of that impulse.

(b) Write down the FIR of $\delta[n - n_0]$ explicitly in the form of Eq. *.

Now further assume that the system is linear. In other words, the response to a linear combination of input is the same linear combination of the response to each individual input.

(c) Obtain the response of the above $g[n]$ to a linear-time-invariant discrete time system.

(a)

$$\begin{aligned}g[n] &= \sum_{m=-\infty}^{\infty} g[m]\delta[n-m] \\&= g[2]\delta[n-2] + g[3]\delta[n-3] + g[4]\delta[n-4] + g[5]\delta[n-5] \\&= \delta[n-2] + \delta[n-3] + 2\delta[n-4] + 2\delta[n-5]\end{aligned}$$

(b)

$$h[n - n_0] = \begin{cases} 1 & n = n_0, n_0 + 1, n_0 + 2 \\ 0 & \text{Otherwise} \end{cases}$$

(c)

$$h[n-2] + h[n-3] + 2h[n-4] + 2h[n-5] = \begin{cases} 1 + 0 + 0 + 0 = 1 & n = 2 \\ 1 + 1 + 0 + 0 = 2 & n = 3 \\ 1 + 1 + 2 + 0 = 4 & n = 4 \\ 0 + 1 + 2 + 2 = 5 & n = 5 \\ 0 + 0 + 2 + 2 = 4 & n = 6 \\ 0 + 0 + 0 + 2 = 2 & n = 7 \\ 0 & \text{Otherwise} \end{cases}$$

Consider the signal $h(t) = 2 \sin(\pi t) - \sin(5\pi t / 2)$.

It is given that
$$\sin \theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}.$$

The (continuous) Fourier transform of $h(t)$ is given by

$$H(f) = \int_{-\infty}^{\infty} h(t) e^{2\pi i f t} dt.$$

Besides, we have
$$\int_{-\infty}^{\infty} e^{2\pi i (f-f')t} dt = \delta(f - f').$$

(a) Sketch the power spectrum, viz., $|H(f)|^2$ vs f , of the above signal $h(t)$.

The signal is sampled at $t = 0, 1, 2, 3$. The four sampled data are denoted by

$$h_0 = h(0), h_1 = h(1), h_2 = h(2), h_3 = h(3).$$

(b) Write down the values of h_0, h_1, h_2, h_3 .

(c) What is the Nyquist critical frequency f_c ?

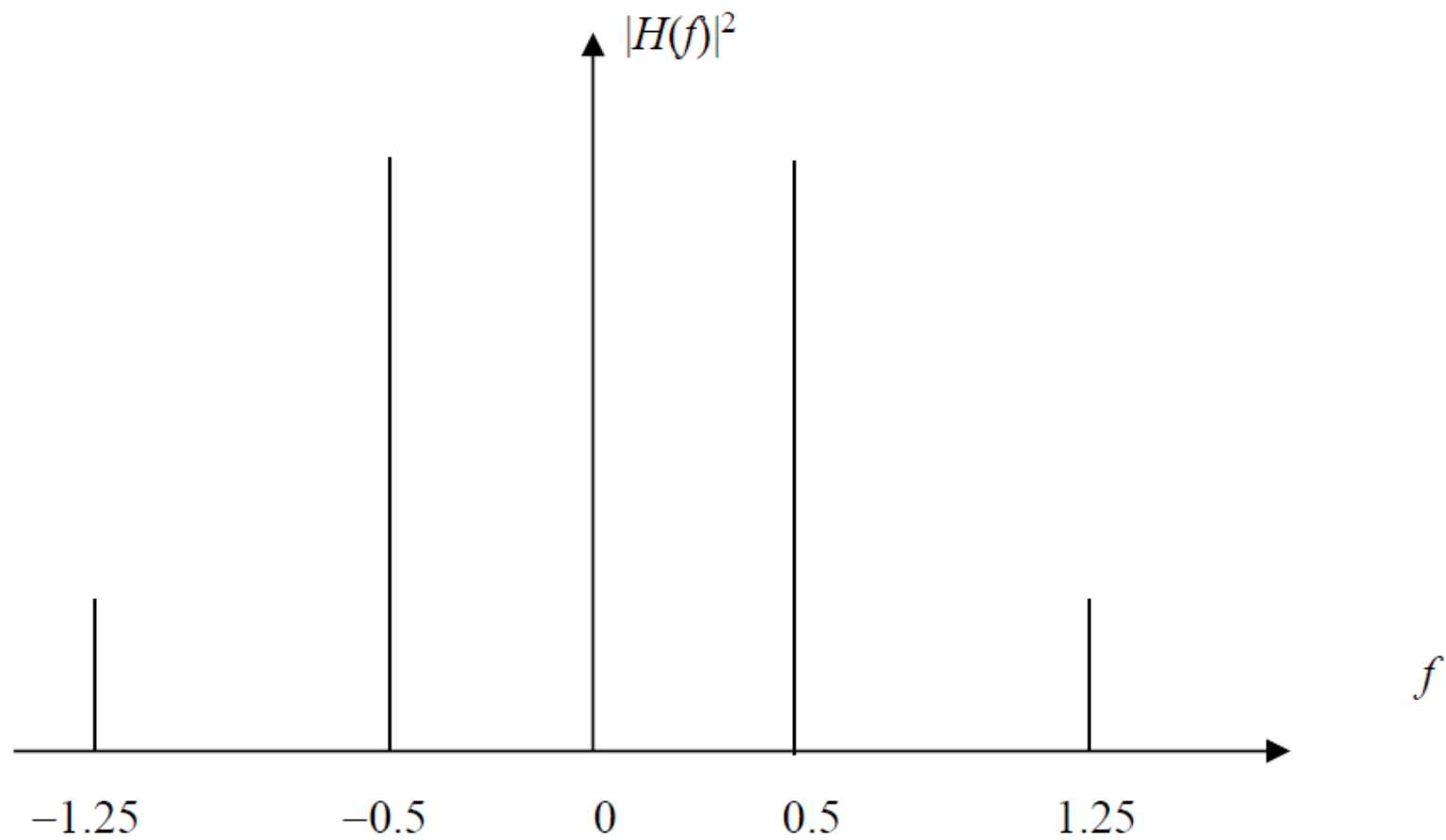
The discrete Fourier transform (DFT) of h_k ($k = 0, 1, 2, \dots, N-1$) is given by

$$H_n = \sum_{k=0}^{N-1} h_k e^{2\pi i k n / N}$$

(d) Obtain H_0, H_1, H_2, H_3 from the above formula. Also write down the corresponding frequencies of the DFT inside the range $-f_c < f \leq f_c$.

(e) Account for the value of H_2 you obtained in terms of aliasing.

(a) $H(f) = -i\delta(f + 0.5) + i\delta(f - 0.5) + \frac{i}{2}\delta(f + 1.25) - \frac{i}{2}\delta(f - 1.25)$



(b) $h_0 = 0, \quad h_1 = -1, \quad h_2 = 0, \quad h_3 = 1.$

$$(c) \quad f_c = \frac{1}{2\Delta} = \frac{1}{2 \times 1} = 0.5$$

(d) $N = 4$.

$$H_0 = \sum_{k=0}^3 h_k = 0, \text{ at } f_0 = \frac{0}{4 \times 1} = 0.$$

$$H_1 = \sum_{k=0}^3 h_k e^{2\pi i k/4} = (-1)e^{2\pi i(1)/4} + (1)e^{2\pi i(3)/4} = -2i, \text{ at } f_1 = \frac{1}{4 \times 1} = 0.25.$$

$$H_2 = \sum_{k=0}^3 h_k e^{2\pi i k(2)/4} = (-1)e^{2\pi i(1)(2)/4} + (1)e^{2\pi i(3)(2)/4} = 0, \text{ at } f_2 = \frac{2}{4 \times 1} = 0.5.$$

$$H_3 = \sum_{k=0}^3 h_k e^{2\pi i k(3)/4} = (-1)e^{2\pi i(1)(3)/4} + (1)e^{2\pi i(3)(3)/4} = 2i, \text{ at } f_3 = \frac{3}{4 \times 1} = 0.75, \text{ which is}$$

shifted to -0.25 inside the range $-f_c < f \leq f_c$.

- (e) $H_2 = 0$, despite that the original signal has the component $2 \sin(\pi t)$, which has a frequency of 0.5. The reason is because the sine wave in fact consists of two frequencies, +0.5 and -0.5. Due to aliasing, the Fourier transforms of these two components add and gives zero, because they are two complex numbers differ by a minus sign.

Consider the probability distribution function (PDF)

$$p(x) = Ax \cos \frac{\pi x}{2} \quad x \in [0,1]$$

- (a) Show that $A = \frac{\pi^2}{2(\pi-2)}$.
- (b) Explain why the simple transformation method fails to sample the above PDF.
- (c) Sample the distribution $p(x)$ by rejection method using a suitable comparison function $f(x)$. Show the steps clearly and find the rate of acceptance. Your score in this part will be correlated to the efficiency of your scheme. Therefore, try to choose reasonably good $f(x)$. If you choose a trivial $f(x)$ like $f(x) = \text{Constant}$, your result will be accepted but the score will be low. In addition, you need to give rigorous proof or arguments to show that the $f(x)$ you choose is a valid comparison function.

(a)

$$\begin{aligned}\int_0^1 Ax \cos \frac{\pi x}{2} dx &= A \frac{2}{\pi} \int_0^1 x d \left(\sin \frac{\pi x}{2} \right) = A \frac{2}{\pi} \left(x \sin \frac{\pi x}{2} \Big|_0^1 - \int_0^1 \sin \frac{\pi x}{2} dx \right) \\ &= A \frac{2}{\pi} \left(1 - \int_0^1 \sin \frac{\pi x}{2} dx \right) = A \frac{2}{\pi} \left(1 + \frac{2}{\pi} \cos \frac{\pi x}{2} \Big|_0^1 \right) = A \frac{2}{\pi} \left(1 - \frac{2}{\pi} \right) = 1\end{aligned}$$

$$A = \frac{\pi}{2} \frac{\pi}{\pi - 2} = \frac{\pi^2}{2(\pi - 2)}$$

(b)

Define

$$\begin{aligned}y = F(x) &= \int_0^x \frac{\pi^2}{2(\pi-2)} x \cos \frac{\pi x}{2} dx = \frac{\pi^2}{2(\pi-2)} \frac{2}{\pi} \int_0^x x d\left(\sin \frac{\pi x}{2}\right) \\&= \frac{\pi}{\pi-2} \left(x \sin \frac{\pi x}{2} \Big|_0^x - \int_0^x \sin \frac{\pi x}{2} dx \right) \\&= \frac{\pi}{\pi-2} \left(x \sin \frac{\pi x}{2} + \frac{2}{\pi} \cos \frac{\pi x}{2} \Big|_0^x \right) \\&= \frac{\pi}{\pi-2} \left(x \sin \frac{\pi x}{2} + \frac{2}{\pi} \cos \frac{\pi x}{2} - \frac{2}{\pi} \right)\end{aligned}$$

The PDF is computable but not invertible.

$$(c) \quad p(x) = \frac{\pi^2}{2(\pi-2)} x \cos \frac{\pi x}{2} \quad x \in [0, 1]$$

Suggested solution 1:

Because $0 \leq x \leq 1$ and $0 \leq \cos \frac{\pi x}{2} \leq 1$,

$$p(x) \leq \frac{\pi^2}{2(\pi-2)} = f(x)$$

Step 1: Generate uniform deviate y from 0 to 1.

Step 2: Generate a second uniform deviate z from 0 to 1.

Step 3: If $z < \frac{p(x)}{\frac{\pi^2}{2(\pi-2)}} = \frac{2(\pi-2)}{\pi^2} p(x)$, return x .

Repeat the above steps for many times.

The area under $f(x)$ is $\frac{\pi^2}{2(\pi-2)}$.

The acceptance rate is $\frac{1}{\frac{\pi^2}{2(\pi-2)}} = \frac{2(\pi-2)}{\pi^2} \approx 0.231$.

$$p(x) = \frac{\pi^2}{2(\pi-2)} x \cos \frac{\pi x}{2} \quad x \in [0,1]$$

Suggested solution 2:

Because $0 \leq x \leq 1$,

$$p(x) \leq \frac{\pi^2}{2(\pi-2)} \cos \frac{\pi x}{2} = f(x).$$

$$\text{Define } y = F(x) = \int_0^x \frac{\pi^2}{2(\pi-2)} \cos \frac{\pi x}{2} dx = \frac{\pi}{\pi-2} \sin \frac{\pi x}{2}.$$

$$\text{Then } x = \frac{2}{\pi} \sin^{-1} \left(\frac{\pi-2}{\pi} y \right).$$

Step 1: Generate uniform deviate y from 0 to $F(1) = \frac{\pi}{\pi-2}$.

Step 2: Evaluate $x = \frac{2}{\pi} \sin^{-1} \left(\frac{\pi-2}{\pi} y \right)$.

Step 3: Generate a second uniform deviate z from 0 to 1.

Step 4: If $z < \frac{p(x)}{f(x)} = x$, return x .

Repeat the above steps for many times.

The area under $f(x)$ is $F(1) = \frac{\pi}{\pi-2}$.

The acceptance rate is $\frac{1}{\frac{\pi}{\pi-2}} = \frac{\pi-2}{\pi} \approx 0.363$.

$$p(x) = \frac{\pi^2}{2(\pi-2)} x \cos \frac{\pi x}{2} \quad x \in [0,1]$$

Suggested solution 3:

Because for $0 \leq x \leq 1$, $0 \leq \cos \frac{\pi x}{2} \leq 1$,

$$p(x) \leq \frac{\pi^2}{2(\pi-2)} x = f(x).$$

Define $y = F(x) = \int_0^x \frac{\pi^2}{2(\pi-2)} x dx = \frac{\pi^2}{4(\pi-2)} x^2$.

$$\text{Then } x = \frac{2\sqrt{\pi-2}}{\pi} \sqrt{y}.$$

Step 1: Generate uniform deviate y from 0 to $F(1) = \frac{\pi^2}{4(\pi-2)}$.

Step 2: Evaluate $x = \frac{2\sqrt{\pi-2}}{\pi} \sqrt{y}$

Step 3: Generate a second uniform deviate z from 0 to 1.

Step 4: If $z < \frac{p(x)}{f(x)} = \cos \frac{\pi x}{2}$, return x .

Repeat the above steps for many times.

The area under $f(x)$ is $F(1) = \frac{\pi^2}{4(\pi-2)}$.

The acceptance rate is $\frac{1}{\frac{\pi^2}{4(\pi-2)}} = \frac{4(\pi-2)}{\pi^2} \approx 0.463$.

$$p(x) = \frac{\pi^2}{2(\pi-2)} x \cos \frac{\pi x}{2} \quad x \in [0,1]$$

Suggested solution 4:

Since $0 \leq x \leq 1$ and cosine is a monotonically decreasing function from 0 to $\pi/2$,

$x^2 < x$ and hence $\cos \frac{\pi x}{2} \leq \cos \frac{\pi x^2}{2}$. Therefore,

$$p(x) \leq \frac{\pi^2}{2(\pi-2)} x \cos \frac{\pi x^2}{2} = f(x).$$

$$\text{Define } y = F(x) = \int_0^x \frac{\pi^2}{2(\pi-2)} x \cos \frac{\pi x^2}{2} dx = \frac{\pi}{2(\pi-2)} \sin \frac{\pi x^2}{2}.$$

$$\text{Then } x = \sqrt{\frac{2}{\pi} \sin^{-1} \left[\frac{2(\pi-2)}{\pi} y \right]}.$$

$$\text{Step 1: Generate uniform deviate } y \text{ from 0 to } F(1) = \frac{\pi}{2(\pi-2)}.$$

$$\text{Step 2: Evaluate } x = \sqrt{\frac{2}{\pi} \sin^{-1} \left[\frac{2(\pi-2)}{\pi} y \right]}.$$

Step 3: Generate a second uniform deviate z from 0 to 1.

$$\text{Step 4: If } z < \frac{p(x)}{f(x)} = \frac{\cos \frac{\pi x}{2}}{\cos \frac{\pi x^2}{2}}, \text{ return } x.$$

Repeat the above steps for many times.

$$\text{The area under } f(x) \text{ is } F(1) = \frac{\pi}{2(\pi-2)} \approx 1.376.$$

$$\text{The acceptance rate is } \frac{1}{F(1)} = \frac{2(\pi-2)}{\pi} \approx 0.727.$$

- (a) Use integration by parts to show that

$$\int x \sin x dx = \sin x - x \cos x + C .$$

- (b) Consider the probability distribution function

$$p(x) = \begin{cases} Ax \sin x & -\pi/4 \leq x \leq \pi/4 \\ 0 & \text{Otherwise} \end{cases}$$

Find A .

- (c) State why one cannot use the transformation method to sample $p(x)$.

- (d) Argue that

$$f(x) = \begin{cases} A|x| & -\pi/4 \leq x \leq \pi/4 \\ 0 & \text{Otherwise} \end{cases}$$

and

$$f(x) = \begin{cases} A|\sin x| & -\pi/4 \leq x \leq \pi/4 \\ 0 & \text{Otherwise} \end{cases}$$

are both valid comparison functions for using rejection method.

- (e) Sample $p(x)$ by the rejection method. You are required to choose one of the two comparison functions in (d) so that the rejection rate is less than 1/2. State the steps clearly.

$$(a) \quad \int x \sin x dx = -\int x d(\cos x) = -x \cos x + \int \cos x dx = \sin x - x \cos x + C$$

(b)

$$\int_{-\pi/4}^{\pi/4} Ax \sin x dx = 1$$

$$A \left[\sin x - x \cos x \right]_{-\pi/4}^{\pi/4} = 1$$

$$A \left[\left(\sin \frac{\pi}{4} - \frac{\pi}{4} \cos \frac{\pi}{4} \right) - \left(\sin \left(-\frac{\pi}{4} \right) - \left(-\frac{\pi}{4} \right) \cos \left(-\frac{\pi}{4} \right) \right) \right] = 1$$

$$A \left[\left(\frac{1}{\sqrt{2}} - \frac{\pi}{4} \frac{1}{\sqrt{2}} \right) - \left(-\frac{1}{\sqrt{2}} + \frac{\pi}{4} \frac{1}{\sqrt{2}} \right) \right] = 1$$

$$\frac{A}{\sqrt{2}} \left[\left(1 - \frac{\pi}{4} \right) - \left(-1 + \frac{\pi}{4} \right) \right] = 1$$

$$A = \frac{2\sqrt{2}}{4 - \pi}$$

$$(c) \quad \int x \sin x dx = -\int x d(\cos x) = -x \cos x + \int \cos x dx = \sin x - x \cos x + C$$

The PDF is computable but not invertible.

(d) The PDF can be written as

$$p(x) = \begin{cases} A|x||\sin x| & -\pi/4 \leq x \leq \pi/4 \\ 0 & \text{Otherwise} \end{cases}$$

Because $|\sin x| \leq 1$ and for $-\pi/4 \leq x \leq \pi/4$, $|x| \leq 1$. Hence $f(x) \geq p(x)$.

(e) Because $\int_{-\pi/4}^{\pi/4} A|x|dx = 2A \int_0^{\pi/4} xdx = A[x^2]_0^{\pi/4} = \frac{2\sqrt{2}}{4-\pi} \left(\frac{\pi}{4}\right)^2 > 2$, this comparison

function has a rejection rate which is greater than 1/2.

On the other hand,

$$\int_{-\pi/4}^{\pi/4} A|\sin x|dx = 2A \int_0^{\pi/4} \sin xdx = 2A[-\cos x]_0^{\pi/4} = \frac{4\sqrt{2}}{4-\pi} \left(1 - \frac{1}{\sqrt{2}}\right) < 2.$$

Hence we should choose $f(x) = \begin{cases} A|\sin x| & -\pi/4 \leq x \leq \pi/4 \\ 0 & \text{Otherwise} \end{cases}$

Let $y = F(x) = \int_{-\pi/4}^x f(x)dx$.

For $x \leq 0$,

$$y = F(x) = \int_{-\pi/4}^x A|\sin x|dx = \int_{-\pi/4}^x A(-\sin x)dx = A[\cos x]_{-\pi/4}^x = A\left(\cos x - \frac{1}{\sqrt{2}}\right)$$

For $x > 0$,

$$\begin{aligned} y = F(x) &= \int_{-\pi/4}^x A|\sin x|dx = \int_{-\pi/4}^0 A|\sin x|dx + \int_0^x A|\sin x|dx \\ &= A[\cos x]_{-\pi/4}^0 + \int_0^x A \sin xdx \\ &= A\left(1 - \frac{1}{\sqrt{2}}\right) + A[-\cos x]_0^x \\ &= A\left(1 - \frac{1}{\sqrt{2}}\right) + A(1 - \cos x) \\ &= A\left(2 - \frac{1}{\sqrt{2}} - \cos x\right) \end{aligned}$$

In other words,

$$y = F(x) = \begin{cases} A\left(\cos x - \frac{1}{\sqrt{2}}\right) & x \leq 0 \\ A\left(2 - \frac{1}{\sqrt{2}} - \cos x\right) & x > 0 \end{cases}$$

$$x = F^{-1}(y) = \begin{cases} \cos^{-1}\left(\frac{y}{A} + \frac{1}{\sqrt{2}}\right) & 0 \leq y \leq A\left(1 - \frac{1}{\sqrt{2}}\right) \\ \cos^{-1}\left(2 - \frac{1}{\sqrt{2}} - \frac{y}{A}\right) & A\left(1 - \frac{1}{\sqrt{2}}\right) < y \leq A(2 - \sqrt{2}) \end{cases}$$

Step 1: Generate uniform deviate y from 0 to $A(2 - \sqrt{2})$.

Step 2: If $A\left(1 - \frac{1}{\sqrt{2}}\right) < y \leq A(2 - \sqrt{2})$, set $x = \cos^{-1}\left(2 - \frac{1}{\sqrt{2}} - \frac{y}{A}\right)$.

If $0 \leq y \leq A\left(1 - \frac{1}{\sqrt{2}}\right)$, set $x = \cos^{-1}\left(\frac{y}{A} + \frac{1}{\sqrt{2}}\right)$.

Step 3: Generate a second uniform deviate s from 0 to $A|\sin x|$.

Step 4: If $s > Ax \sin x$, reject the point.

If $s \leq Ax \sin x$, return x .

Step 5: Repeat for many times.

Consider N random deviates $x_1, x_2, \dots, x_N$ with the same probability distribution function (PDF)

$$P(x_i) = \begin{cases} C & 0 \leq x_i < 2 \\ 0 & \text{Otherwise} \end{cases}$$

where C is a constant.

- (a) Show that $C = 1/2$.
- (b) Show that the mean $\langle x_i \rangle = 1$ and the s.d. $\sigma(x_i) = 1/\sqrt{3}$.
- (c) Consider the sum $s = \sum_{i=1}^N x_i$, what approximately is the PDF $P(s)$ when N is very large?

Now consider the case $N = 2$, where $s = x_1 + x_2$.

(d) Show that
$$P(s) = \begin{cases} s/4 & 0 \leq s < 2 \\ 1-s/4 & 2 \leq s < 4 \end{cases}$$

The above result shows that one can generate $P(s)$ by the “direct method” using two uniform deviates from 0 to 2.

- (e) Describe how you generate $P(s)$ by the direct method using a random number generator which returns uniform deviate from 0 to 1.

The same PDF can also be generated using the transformation method.

- (f) Describe how you generate the $P(s)$ in (d) by transformation method. Write down the steps clearly.

Consider the PDF

$$Q(s) = \begin{cases} As(4-s) & 0 \leq s < 4 \\ 0 & \text{Otherwise} \end{cases}$$

- (g) Show that $A = 3/32$.
- (h) Suppose we do not know how to find the roots of cubic equations, explain why one cannot use the transformation method to sample $Q(s)$ then.

Let us sample $Q(s)$ by the rejection method using the comparison function

$$f(s) = \begin{cases} Bs/4 & 0 \leq s < 2 \\ B(1-s/4) & 2 \leq s < 4 \end{cases}$$

where B is a constant.

- (i) Show that the optimal value of B is $3/2$. What is the rejection rate for this choice of B ?
- (j) Describe how you sample $Q(s)$ with the comparison function in (i). Write down the steps clearly.

(a) By normalization, $\int_0^2 P(x_i) dx_i = \int_0^2 C dx_i = 2C = 1 \Rightarrow C = 1/2.$

$$(b) \quad \langle x_i \rangle = \int_0^2 x_i P(x_i) dx_i = \int_0^2 x_i \frac{1}{2} dx_i = \left[\frac{x_i^2}{4} \right]_0^2 = 1.$$

$$\langle x_i^2 \rangle = \int_0^2 x_i^2 P(x_i) dx_i = \int_0^2 x_i^2 \frac{1}{2} dx_i = \left[\frac{x_i^3}{6} \right]_0^2 = \frac{4}{3}.$$

$$\sigma(x_i) = \sqrt{\langle x_i^2 \rangle - \langle x_i \rangle^2} = \sqrt{\frac{4}{3} - 1^2} = \frac{1}{\sqrt{3}}.$$

- (c) By central limit theorem, the PDF should approach a Gaussian distribution. The mean is $N\langle x_i \rangle = N$ and the s.d. is $\sqrt{N}\sigma(x_i) = \sqrt{\frac{N}{3}}$.

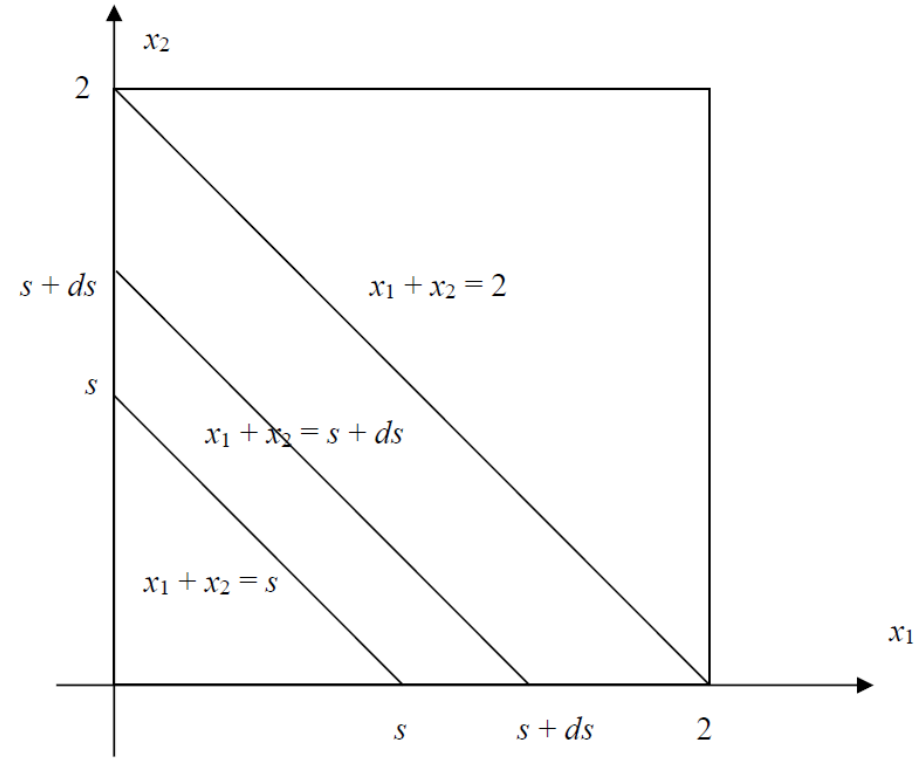
$$(d) \quad P(s) = \int_0^2 P(x_1)P(s-x_1)dx_1.$$

$$\text{For } 0 \leq s < 2, \quad P(s) = \int_0^s P(x_1)P(s-x_1)dx_1 = \frac{1}{2} \times \frac{1}{2} \int_0^s dx_1 = \frac{s}{4}.$$

For $2 \leq s < 4$,

$$P(s) = \int_{s-2}^2 P(x_1)P(s-x_1)dx_1 = \frac{1}{2} \times \frac{1}{2} \int_{s-2}^2 dx_1 = \frac{4-s}{4} = 1 - \frac{s}{4}.$$

One can also derive the result geometrically. Generate x_1 and x_2 and plot the points (x_1, x_2) on the x_1 - x_2 plane. The points will have uniform probability distribution in the square $[0, 2) \otimes [0, 2)$. Draw the line $x_1 + x_2 = s$. The figure below shows the lines $x_1 + x_2 = s$ and $x_1 + x_2 = s + ds$, where we've assumed $s < 2$.



It is easy to show that the area bounded by the two lines is sds . Hence the probability of having the sum between s and $s + ds$ is

$$P(s)ds = \frac{sds}{4} \Rightarrow P(s) = \frac{s}{4}.$$

The probability density for $4 > s > 2$ can be obtained similarly, or by noticing from the above figure that $P(s)$ should be symmetric about $s = 2$, and hence

$$P(s) = 1 - \frac{s}{4} \quad \text{for } 4 > s > 2.$$

- (e) Call the random number generator to obtain a uniform deviate from 0 to 1. Multiply it by 2. Repeat the above once. Return their sum.

(f) Define $t(s) = \int_0^s P(s) ds$.

$$t(2) = 1/2.$$

$$\text{For } s \leq 2, \quad t(s) = \int_0^s \frac{s}{4} ds = \frac{s^2}{8}.$$

$$\text{For } 4 \geq s \geq 2, \quad t(s) = t(2) + \int_2^s \left(1 - \frac{s}{4}\right) ds = \frac{1}{2} + \left[s - \frac{s^2}{8}\right]_2^s = -1 + s - \frac{s^2}{8}.$$

The inverse function is $s(t) = \sqrt{8t}$.

The inverse function is $s(t) = 4 \pm \sqrt{8(1-t)}$.

Because $s = 2$ when $t = 1/2$, we should choose the minus sign.

Therefore,

$$s(t) = \begin{cases} \sqrt{8t} & 0 \leq t < 1/2 \\ 4 - \sqrt{8(1-t)} & 1/2 \leq t \leq 1 \end{cases}$$

Step 1: Generate uniform deviate t from 0 to 1.

Step 2: If $t < 1/2$, return $\sqrt{8t}$.

If $t \geq 1/2$, return $4 - \sqrt{8(1-t)}$.

(g) By normalization,

$$\int_0^4 Q(s) dx = \int_0^4 As(4-s) dx = A \left[2s^2 - \frac{s^3}{3} \right]_0^4 = \frac{32}{3} A = 1 \Rightarrow A = \frac{3}{32}.$$

- (h) $Q(s)$ is computable but the result is a cubic equation. Not knowing how to solve cubic equations, the result is not invertible. Therefore transformation method cannot be used.

(i) Rigorous proof is not required.

By sketching the two functions, one can see a necessary and sufficient condition for $f(s)$ to be always greater than $Q(s)$ is that its derivative at $s = 0$ must not be smaller than that of $Q(s)$. Besides, it is obvious that the area under $f(s)$ and the slope are both proportional to B . Therefore the optimal value of B is such that its slope equals that of $Q(s)$ at $s = 0$:

$$f'(0) = \frac{B}{4} = Q'(0) = \frac{d}{ds} \left[\frac{3}{32} s(4-s) \right] \Big|_{s=0} = \frac{3}{8} \Rightarrow B = \frac{3}{2}.$$

The area under $f(s)$ is $B = 3/2$. Hence the rejection rate is $\frac{3/2 - 1}{3/2} = \frac{1}{3}$.

(j) Define $t(s) = \int_0^s f(s) ds$ and then find the inverse function.

The result is the same as (f), only that t is now scaled by $3/2$. Hence

$$\begin{aligned} s(t) &= \begin{cases} \sqrt{8t \times \frac{2}{3}} & 0 \leq t < 3/4 \\ 4 - \sqrt{8 \left(1 - t \times \frac{2}{3}\right)} & 3/4 \leq t \leq 3/2 \end{cases} \\ &= \begin{cases} \sqrt{16t/3} & 0 \leq t < 3/4 \\ 4 - \sqrt{8(1 - 2t/3)} & 3/4 \leq t \leq 3/2 \end{cases} \end{aligned}$$

Step 1: Generate uniform deviate t from 0 to $3/2$.

Step 2: If $t < 3/4$, evaluate $s = \sqrt{16t/3}$.

If $t \geq 3/4$, evaluate $s = 4 - \sqrt{8(1 - 2t/3)}$.

Step 3: Generate a second uniform deviate y from 0 to 1.

Step 4: If $y < Q(s)/f(s)$, return s ; else reject s and repeat.

Consider the cubic approximation $g(x)$ of the function $f(x)$

$$f(x) \approx g(x) = Ax^3 + Bx^2 + Cx + D$$

which satisfies

$$\begin{cases} g\left(-\frac{3h}{2}\right) = f\left(-\frac{3h}{2}\right) \\ g\left(-\frac{h}{2}\right) = f\left(-\frac{h}{2}\right) \\ g\left(\frac{h}{2}\right) = f\left(\frac{h}{2}\right) \\ g\left(\frac{3h}{2}\right) = f\left(\frac{3h}{2}\right) \end{cases}$$

(a) Find B and D in terms of h , $f\left(-\frac{3h}{2}\right)$, $f\left(-\frac{h}{2}\right)$, $f\left(\frac{h}{2}\right)$ and $f\left(\frac{3h}{2}\right)$.

(b) Obtain the formula of the approximation of

$$\int_{-3h/2}^{3h/2} f(x)dx \approx \int_{-3h/2}^{3h/2} (Ax^3 + Bx^2 + Cx + D)dx$$

in terms of h , $f\left(-\frac{3h}{2}\right)$, $f\left(-\frac{h}{2}\right)$, $f\left(\frac{h}{2}\right)$ and $f\left(\frac{3h}{2}\right)$.

(c) Obtain the composite rule for

$$\int_a^b f(x)dx$$

in terms of h and $f(x_k)$, where $x_k = a + kh$, $k = 0, 1, 2, \dots, N$, and $h = \frac{b-a}{N}$ with N being a multiple of 3.

By substitution,

(a)

$$\begin{aligned}f\left(-\frac{3h}{2}\right) &= g\left(-\frac{3h}{2}\right) = A\left(-\frac{3h}{2}\right)^3 + B\left(-\frac{3h}{2}\right)^2 + C\left(-\frac{3h}{2}\right) + D \\f\left(-\frac{h}{2}\right) &= g\left(-\frac{h}{2}\right) = A\left(-\frac{h}{2}\right)^3 + B\left(-\frac{h}{2}\right)^2 + C\left(-\frac{h}{2}\right) + D \\f\left(\frac{h}{2}\right) &= g\left(\frac{h}{2}\right) = A\left(\frac{h}{2}\right)^3 + B\left(\frac{h}{2}\right)^2 + C\left(\frac{h}{2}\right) + D \\f\left(\frac{3h}{2}\right) &= g\left(\frac{3h}{2}\right) = A\left(\frac{3h}{2}\right)^3 + B\left(\frac{3h}{2}\right)^2 + C\left(\frac{3h}{2}\right) + D\end{aligned}$$

Summing 1st and 4th, 2nd and 3rd equation gives

$$\begin{aligned}f\left(-\frac{3h}{2}\right) + f\left(\frac{3h}{2}\right) &= 2B\left(\frac{3h}{2}\right)^2 + 2D \\f\left(-\frac{h}{2}\right) + f\left(\frac{h}{2}\right) &= 2B\left(\frac{h}{2}\right)^2 + 2D\end{aligned}$$

Therefore

$$\begin{aligned}B &= \frac{1}{4h^2} \left[f\left(-\frac{3h}{2}\right) + f\left(\frac{3h}{2}\right) - f\left(-\frac{h}{2}\right) - f\left(\frac{h}{2}\right) \right] \\D &= \frac{1}{16} \left[9f\left(-\frac{h}{2}\right) + 9f\left(\frac{h}{2}\right) - f\left(-\frac{3h}{2}\right) - f\left(\frac{3h}{2}\right) \right]\end{aligned}$$

(b)

$$\begin{aligned}\int_{-3h/2}^{3h/2} f(x) dx &\approx \int_{-3h/2}^{3h/2} (Ax^3 + Bx^2 + Cx + D) dx \\&= \left[\frac{Ax^4}{4} + \frac{Bx^3}{3} + \frac{Cx^2}{2} + Dx \right]_{-3h/2}^{3h/2} \\&= \frac{2B}{3} \left(\frac{3h}{2} \right)^3 + 2D \left(\frac{3h}{2} \right) \\&= \frac{9h}{16} \left[f\left(-\frac{3h}{2}\right) + f\left(\frac{3h}{2}\right) - f\left(-\frac{h}{2}\right) - f\left(\frac{h}{2}\right) \right] \\&\quad + \frac{3h}{16} \left[9f\left(-\frac{h}{2}\right) + 9f\left(\frac{h}{2}\right) - f\left(-\frac{3h}{2}\right) - f\left(\frac{3h}{2}\right) \right] \\&= \frac{3h}{8} \left[f\left(-\frac{3h}{2}\right) + 3f\left(-\frac{h}{2}\right) + 3f\left(\frac{h}{2}\right) + f\left(\frac{3h}{2}\right) \right]\end{aligned}$$

(c)

The total area under a cubic curve:

$$\begin{aligned}\int_a^b f(x)dx &\approx \frac{3h}{8} [f(a) + 3f(a+h) + 3f(a+2h) + f(a+3h)] \\ &\quad + \frac{3h}{8} [f(a+3h) + 3f(a+4h) + 3f(a+5h) + f(a+6h)] + \dots \\ &= \frac{3h}{8} \left[f(a) + f(b) + 3 \sum_{k=1}^{\frac{N}{3}} f(a + (3k-2)h) + 3 \sum_{k=1}^{\frac{N}{3}} f(a + (3k-1)h) \right. \\ &\quad \left. + 2 \sum_{k=1}^{\frac{N}{3}-1} f(a + 3kh) \right]\end{aligned}$$

The Planck theory of thermal radiation tells us that in the (angular) frequency interval ω to $\omega + d\omega$, a black body of unit area radiates electromagnetically an amount of thermal energy per second equal to $I(\omega) d\omega$, where

$$I(\omega) = \frac{h}{8\pi^3 c^2} \frac{\omega^3}{\left(e^{h\omega/2\pi k_B T} - 1\right)}$$

Here h is the Planck's constant, c is the speed of light, and k_B is Boltzmann's constant. In other words, the total power per unit area radiated by a black body is

$$W = \int_0^\infty I(\omega) d\omega = \int_0^\infty \frac{h}{8\pi^3 c^2} \frac{\omega^3}{\left(e^{h\omega/2\pi k_B T} - 1\right)} d\omega$$

It is obvious from the above integral that W depends on T only.

You do not need to understand the physics of blackbody radiation. This question is to serve as an example of how to combine analytical and numerical techniques to obtain the answer.

To find how W depends on T , it is a wrong method to evaluate the integral directly for many times with different values of T and tabulate W and T . Instead, one only needs to evaluate the integral once.

- (a) By performing a change of variable in the integral show that

$$W = \sigma T^4$$

where σ is a fundamental constant (called the Stefan–Boltzmann constant) given by

$$\sigma = \frac{2\pi k_B^4}{c^2 h^3} \int_0^\infty \frac{x^3}{e^x - 1} dx$$

- (b) To evaluate the improper integral, perform a change of variable

$$z = \frac{x}{1+x}$$

so that the integral reads

$$I = \int_0^\infty \frac{x^3}{e^x - 1} dx = \int_0^1 f(z) dz$$

Find $f(z)$.

a) By change of variable $x = \frac{h\omega}{2\pi k_B T}$,

$$\begin{aligned} W &= \int_0^\infty \frac{h}{8\pi^3 c^2} \frac{\omega^3}{\left(e^{h\omega/2\pi k_B T} - 1\right)} d\omega \\ &= \int_0^\infty \frac{h}{8\pi^3 c^2} \left(\frac{2\pi k_B T}{h}\right)^4 \frac{x^3}{(e^x - 1)} dx \\ &= \frac{2\pi k_B^4 T^4}{h^3 c^2} \int_0^\infty \frac{x^3}{(e^x - 1)} dx \end{aligned}$$

b) By change of variable $z = \frac{x}{1+x} \Rightarrow x = \frac{z}{1-z}$, $dx = \frac{dz}{(1-z)^2}$,

$$\int_0^{\infty} \frac{x^3}{(e^x - 1)} dx = \int_0^1 \frac{1}{(1-z)^5} \frac{z^3}{\left(e^{\frac{z}{1-z}} - 1\right)} dz$$